

APPLICATIONS

THE LOGARITHMIC GENERALIZED INVERSE GAUSSIAN DISTRIBUTION (LNGIG)

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Key words: Base metal and precious metal ore grade distributions; distributions involving Bessel functions; estimation by method of moments; pseudo lognormal distribution; relationship between skewness and kurtosis; reverse J-shaped; sub lognormal and hyper lognormal.

Summary: Few mathematical models exist for log-transformed data with tails thinner than those of the normal probability law. In this study, the Lognormal Generalised Inverse Gaussian (LNGIG) distribution is developed, based on the logarithmic transformation of the four-parameter Generalised Inverse Gaussian (GIG) distribution. This logarithmic transformation also enables us to use the method of moments for the estimation of parameters. Two applications are given involving the size of oil and gas fields and the distribution of individual ore sample values from a hydrothermal gold deposit.

1. Introduction

There are many frequency distributions of non-negative events (z) which are reverse-J shaped. In these cases, it is customary to carry out a logarithmic transformation of the data in order to obtain a normal distribution and to stabilise variance. However, data obtained thus will not necessarily be normally distributed and may still display positive or negative skewness with heavy or

thin tails. In such cases the original $\Lambda(z)$ distributions are said to be "pseudo-lognormal".

The LNGIG is continuous over the real line with platykurtic or leptokurtic tails. It is called "sub-lognormal" if its tails are short and thin and "hyper-lognormal" if its tails are long and heavy.

Most theoretical probability distributions are leptokurtic with coefficient of kurtosis $\beta_2 \geq 3$. The best known platykurtic distributions (i.e. $\beta_2 < 3$) are the skew Type I and symmetric Type II distributions in the Pearson system. They have finite threshold parameters at both ends of the variate scale, which may lead to difficulties if the method of maximum likelihood estimation for parameters is used. Another awkward problem may arise if, after fitting the observed frequency distribution, new observations outside the termini are encountered.

2. Logarithmic transformation of the generalized inverse Gaussian distribution

The density function of the three-parameter generalised inverse Gaussian (GIG) may be given by

$$\Lambda(z) = \frac{e^{-\gamma\xi}}{2K_\gamma(b)} z^{\gamma-1} e^{-\frac{b}{2}[e^{-\xi}z + e^\xi z^{-1}]} \quad (1)$$

$$0 \leq z < \infty, -\infty < \xi < \infty, -\infty < \gamma < \infty \text{ and } b > 0,$$

where $K_\gamma(b)$ is the modified Bessel function of the second kind of order γ and argument b .

This distribution has a zero ordinate at the origin, rises rather steeply to a maximum and then gently declines to zero with a long, drawn-out tail to the right. In shape it mimics the lognormal distribution. Jörgenson (1982) has extensively researched this distribution from both the theoretical and practical points of view.

As in the case of the lognormal probability law, the transformation $x = \ln z$ is useful for parameter estimation by the method of moments and also for approaching 'normality'. The transformed distribution becomes

$$\varphi(x) = \frac{1}{2K_\gamma(b)} e^{\gamma x - \xi - b \cosh x - \xi} \quad (2)$$

$$-\infty < x < \infty, -\infty < \xi < \infty, -\infty < \gamma < \infty \text{ and } b < 0.$$

Introducing a scale parameter, s , for purposes of practical data analysis, yields the four-parameter LNGIG distribution given by:

$$\varphi(x) = \frac{1}{2sK_\gamma(b)} e^{\gamma \left(\frac{x - \xi}{s} \right) - b \cosh \left(\frac{x - \xi}{s} \right)} \quad (3)$$

$$-\infty < x < \infty, -\infty < \xi < \infty, -\infty < \gamma < \infty, b > 0 \text{ and } s > 0$$

If $\gamma > 0$, the skewness of (3) is negative, if $\gamma < 0$, it has positive skewness and for $\gamma = 0$, it is symmetric. The parameter b determines the kurtosis of the LNGIG distribution. It is always positive but may be arbitrarily small. The LNGIG was used by Sichel, Kleingeld and Assibey-Bonsu (1992) and Dohm (1995) in describing certain observed distributions of individual *in situ* gold ore values.

Letting $u = (x - \xi)/s$ in (3) yields the standard form of the LNGIG distribution:

$$\Omega(u) = \frac{1}{2K_\gamma(b)} e^{\gamma u - b \cosh u} \quad (4)$$

$$-\infty < u < \infty, -\infty < \gamma < \infty \text{ and } b > 0.$$

3. Moments of the standardized LNGIG distribution

The moment generating function associated with (4) is

$$M_u(t) = \frac{K_{\gamma+t}(b)}{K_\gamma(b)}. \quad (5)$$

Successive differentiation of (5) with respect to t gives the first four moments about the origin,

$$\mu'_r(u) = \frac{1}{K_\gamma(b)} \left. \frac{\partial^r K_{\gamma+t}(b)}{\partial t^r} \right|_{t=0} \quad r = 1, 2, 3, 4. \quad (6)$$

The skewness and kurtosis coefficients are given by (7) and (8) respectively:

$$\sqrt{\beta_1(u)} = \sqrt{\beta_1(x)} = \frac{\mu_3(u)}{[\mu_2(u)]^{3/2}} \quad (7)$$

$$\beta_2(u) = \beta_2(x) = \frac{\mu_4(u)}{[\mu_2(u)]^2}. \quad (8)$$

4. The β_2 versus $\sqrt{\beta_1}$ relationship for the LNGIG distribution and the limiting function as $b \rightarrow 0$

A useful device for studying the relationship between β_2 and $\sqrt{\beta_1}$ for a given statistical distribution function is a plot of β_2 against $\sqrt{\beta_1}$. Table 1 provides values of β_2 vs. $-\sqrt{\beta_1}$ for fixed $\gamma = 0.10$ and b ranging from 0 to 20.

This reveals that the distribution approaches the normal for large b . As $-\sqrt{\beta_1}$ increases, β_2 first decreases from $\beta_2 = 3.00$ to $\beta_2 = 2.17$ and then increases to $\beta_2 = 8.83$ as $b \rightarrow 0$ and $-\sqrt{\beta_1} \rightarrow 1.96$. Such behaviour of the relationship between $-\sqrt{\beta_1}$ and β_2 is quite unusual, since for virtually all known statistical distributions the relationship is monotonic.

Table 1: β_2 as a function of $-\sqrt{\beta_1}$, for the LNGIG, with $\gamma = 0.10$ fixed

b	$-\sqrt{\beta_1}$	β_2	b	$-\sqrt{\beta_1}$	β_2
0	1.95958	8.83207	0.010	0.28325	2.10732
10^{-10}	1.38761	4.80064	0.020	0.23692	2.18024
10^{-9}	1.27902	4.32226	0.030	0.21038	2.19439
10^{-8}	1.15973	3.85927	0.040	0.19191	2.20855
10^{-7}	1.03066	3.42531	0.050	0.17784	2.22019
10^{-6}	0.89290	3.03335	0.100	0.13598	2.27857
10^{-5}	0.74769	2.69580	0.500	0.05639	2.49595
10^{-4}	0.59632	2.42570	1.000	0.03314	2.61454
0.001	0.44040	2.24024	5.000	0.00627	2.85363
0.002	0.39297	2.20498	10.0	0.00260	2.91597
0.003	0.36521	2.18960	20.0	0.00101	2.95443
0.005	0.33031	2.17630	∞	0	3

The limiting case of (4) as $b \rightarrow 0$ for given $\gamma > 0$ is the log-gamma density function given by:

$$f(v) = \frac{e^{\gamma v - e^v}}{\Gamma(\gamma)} \quad (9)$$

$$-\infty < v < \infty, \sqrt{\beta_1} < 0, \beta_2 > 3 \text{ and } v = u + \ln(b/2)$$

whose coefficients of skewness and kurtosis are:

$$-\sqrt{\beta_1(v)} = \frac{-\Psi''(\gamma-1)}{[\Psi'(\gamma-1)]^{3/2}} \quad (10)$$

and

$$\beta_2(v) = 3 + \frac{\Psi'''(\gamma - 1)}{[\Psi'(\gamma - 1)]^2}, \quad (11)$$

respectively, where Ψ' , Ψ'' and Ψ''' are the tri-, tetra- and pentagamma functions respectively, defined by

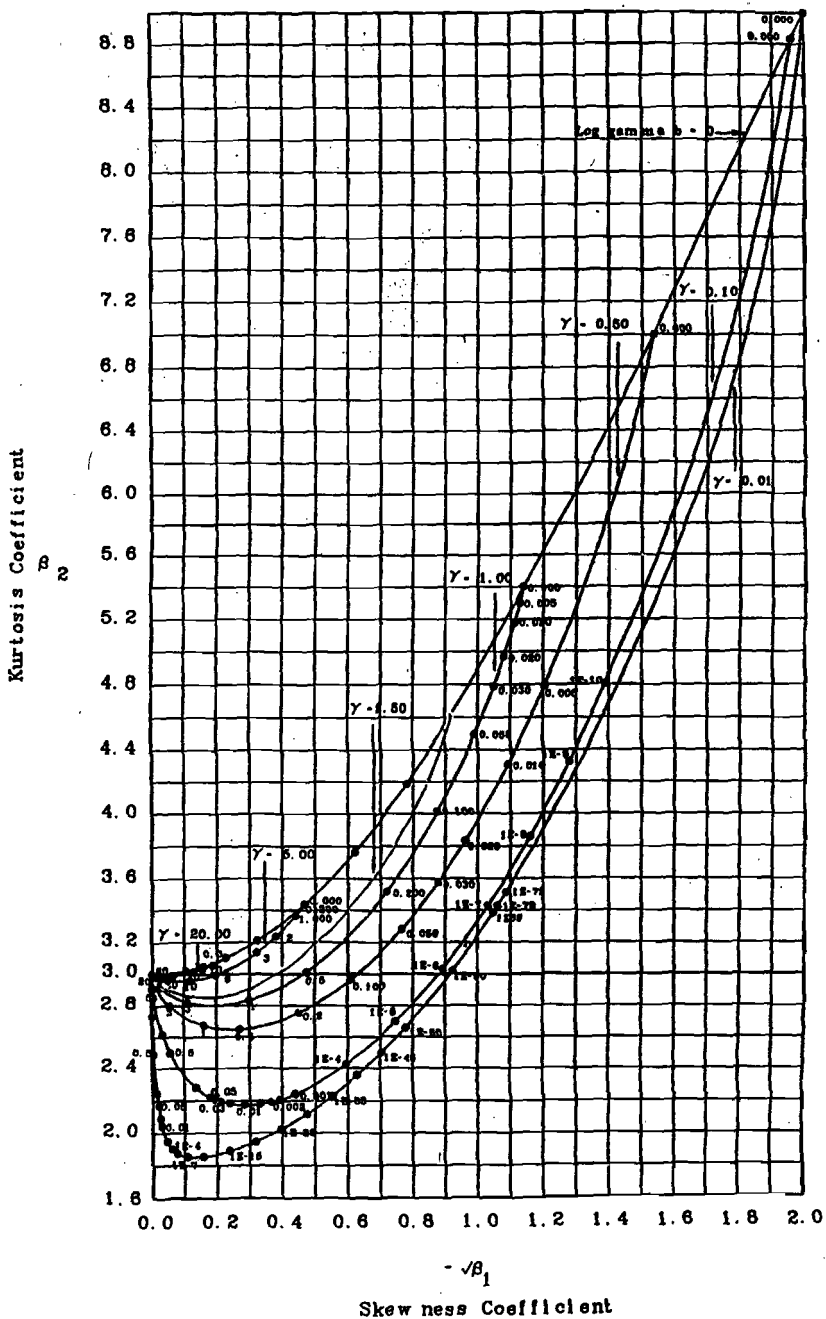
$$\Psi^{(n-1)}(g) = \frac{d^n \ln \Gamma(g+1)}{dg^n} \quad n = 2, 3, 4. \quad (12)$$

Some limiting values, calculated from (10) and (11), are shown in table 2.

Table 2: Limiting the function of β_2 versus $-\sqrt{\beta_1}$, for LNGIG and $b \rightarrow 0$, derived from the log-gamma distribution

γ	$-\sqrt{\beta_1}$	β_2	γ	$-\sqrt{\beta_1}$	β_2
0	2	9	3.00	0.62095	3.76256
0.01	1.99952	8.99805	5.00	0.46859	3.43745
0.10	1.95958	8.83207	10.00	0.32400	3.20976
0.50	1.53514	7.00000	20.00	0.22638	3.10247
1.00	1.13955	5.40000	40.00	0.15910	3.05062
1.50	0.91700	4.61250	90.00	0.10570	3.02234
2.00	0.78024	4.18752	∞	0	3

The LNGIG may be a suitable model for observed pairs of $-\sqrt{\beta_1}$ and β_2 which lie below the values in the above table. On the other hand, any $-\sqrt{\beta_1}$ and β_2 which lie above the limiting curve, indicate that the LNGIG distribution is unsuitable. Figure 1 illustrates this relationship for fixed values of $\gamma = 0.01, 0.10, 0.50, 1.00, 1.50, 5.00$ and 20.00 and b ranging from 0 to 20, as well as the limiting case $b \rightarrow 0$.



5. Parameter estimation for the LNGIG by the method of moments

For an observed pair $(-\sqrt{\beta_1(x)}, \beta_2(x))$ from a given data set, its position in Figure 1 will roughly indicate initial estimates for parameters γ and b . A numerical differentiation computer program can be used to calculate the central moments as well as equations (7) and (8) for any given γ and b .

It should be noted that the shape coefficients for the standardised variable, $u = (x - \xi)/s$ are identical to those of $x = \ln z$, i.e.

$$\sqrt{\beta_1(u)} = \sqrt{\beta_1(x)} \text{ and } \beta_2(u) = \beta_2(x). \quad (13)$$

The resulting numerical values of (7) and (8) for the standardised u -variable can then be compared with the observed skewness and kurtosis of the x -variable. Successive iterations will lead to final estimates for γ and b which match the observed $-\sqrt{\hat{\beta}_1(x)}$ and $\hat{\beta}_2(x)$ values.

To estimate the location and scale parameters, note that from the linear transformation;

$$x = su + \xi$$

it follows that

$$\mu'_1(x) = s\mu'_1(u) + \xi$$

and

$$\mu_2(x) = s^2\mu_2(u),$$

so that

$$s = \sqrt{\frac{\mu_2(x)}{\mu_2(u)}} \quad (14)$$

and

$$\xi = \mu'_1(x) - s\mu'_1(u) \quad (15)$$

where $\hat{\mu}'_1(u)$ and $\hat{\mu}'_2(u)$ are obtained from the numerical solutions of (6) and $\hat{\mu}'_1(x)$ and $\hat{\mu}'_2(x)$ are the mean and variance of the observed distribution respectively. It has been found in practice that these estimators perform almost as well as the maximum likelihood estimators.

6. Modes and modal ordinates for $n\varphi(x)$ and $n\Lambda(z)$

After differentiation of $\ln \varphi(x)$, we find that the mode of the LNGIG distribution is given by

$$u_M = \sin h^{-1} \left(\frac{\gamma}{b} \right) \quad (16)$$

and

$$x_M = \xi + s \cdot \sin h^{-1} \left(\frac{\gamma}{b} \right) \quad (17)$$

The modal ordinate is thus

$$n\varphi(x_M) = \frac{n}{2sK_\gamma(b)} e^{\gamma u_M - b \cosh u_M}, \quad (18)$$

where n is the sample size. In some cases of fitting the LNGIG to observed data, it is required to go back to the untransformed distribution $\Lambda(z)$ via the back-transformation $z = e^x$. This has density function:

$$\Lambda(z) = \frac{1}{2sK_\gamma(b)z} e^{\gamma \left(\frac{\ln z - \xi}{s} \right) - b \cosh \left(\frac{\ln z - \xi}{s} \right)}$$

$$0 \leq z < \infty, \quad -\infty < \xi < \infty, \quad -\infty < \gamma < \infty, \quad b > 0$$

$$\text{and } s > 0. \quad (19)$$

Differentiation gives the mode of $\Lambda(z)$ as

$$u'_M = \sin h^{-1} \left(\frac{\gamma - s}{b} \right) \quad (20)$$

and

$$z_M = e^{\xi + s \sin h^{-1} \left(\frac{\gamma - s}{b} \right)} \quad (21)$$

The modal ordinate is thus

$$n\Lambda(z_M) = \frac{ne^{-\xi}}{2sK_\gamma(b)} e^{(\gamma-s)u'_M - b \cosh u'_M} \quad (22)$$

7. Two examples of fitting the LNGIG distribution model to observed data

The LNGIG distribution may be fitted, inter alia, to

- (i) The distribution of examination marks.
- (ii) The claim size distribution for short-term car insurance against fire and storm damage, where the logarithm of the claim is employed as the random variable.
- (iii) The distribution of the logarithms of individual ore values, particularly where the geological process of ore formation was of a hydrothermal character.
- (iv) Mixtures of normal distributions with shifting mean values.

7.1 Sizes of oil and gas fields

The first example is taken from an investigation by Coburn and Schuenemeyer (1991) dealing with the sizes of oil and gas fields measured in millions of barrels of oil equivalent (MMBOE) in the Gulf of Mexico. The histogram of the 678

fields is shown at the bottom of their paper on page 431, where the variable is $\log(\text{MMBOE})$.

Although these authors suggest fitting the lognormal distribution, inspection of their histogram indicates platykurtosis and negative skewness. This may be explained by the shift in mean values as larger fields were discovered.

The observed frequency distribution is shown in Table 3 and from the first four moments of this distribution one obtains the following estimates:

$$\begin{aligned}\hat{\mu}'_1(x) &= 2.7943316, \\ \hat{\mu}_2(x) &= 2.6798806, \\ \sqrt{\hat{\beta}_1(x)} &= -0.20443 \text{ and} \\ \hat{\beta}_2(x) &= 2.61244.\end{aligned}$$

The distribution is platykurtic with negative skewness. To test whether the observed $\hat{\beta}_2(x)$ coefficient could have arisen from a normal value of $\beta_2 = 3$ by chance alone, we use the asymptotic standard error of $\hat{\beta}_2$ for a normal parent population (Kendall and Stuart (1958)), that is

$$S.E. [\hat{\beta}_2(x)] = \sqrt{\frac{24}{n}} = \sqrt{\frac{24}{678}} = 0.18814$$

from which

$$t = \frac{2.61244 - 3}{0.18814} = -2.06 < -1.645$$

Hence it follows that the observed $\hat{\beta}_2$ coefficient is significantly less than 3 at the 5% level of significance.

Table 3. Observed frequencies of combined oil and gas fields, in MMBOE, in the Gulf of Mexico. (After Coburn Schuenemeyer (1991))

Class limits in MMBOE (z units)	Class limits in $\ln(\text{MMBOE})$ $x = \ln z$ units	Observed frequency $n\hat{\phi}(x)$
0.15625 – 0.31250	-1.8563 – -1.1632	7
0.31250 – 0.62500	-1.1632 – -0.4700	15
0.62500 – 1.25000	-0.4700 – 0.2231	18
1.25000 – 2.50000	0.2231 – 0.9163	53
2.50000 – 5.00000	0.9163 – 1.6094	69
5.00000 – 10.0000	1.6094 – 2.3026	97
10.0000 – 20.0000	2.3026 – 2.9957	104
20.0000 – 40.0000	2.9957 – 3.6889	97
40.0000 – 80.0000	3.6889 – 4.3820	92
80.0000 – 160.000	4.3820 – 5.0752	76
160.000 – 320.000	5.0752 – 5.7683	39
320.000 – 640.000	5.7683 – 6.4615	6
640.000 – 1280.00	6.4615 – 7.1546	5
Total n		678

The log-probability plot of the frequency distribution in Table 3 is plotted in Figure 2. The central part of the plot is linear and the extremes bend upwards away from the line at the upper end and downwards at the lower end, indicating a platykurtic, sub-lognormal distribution for x .

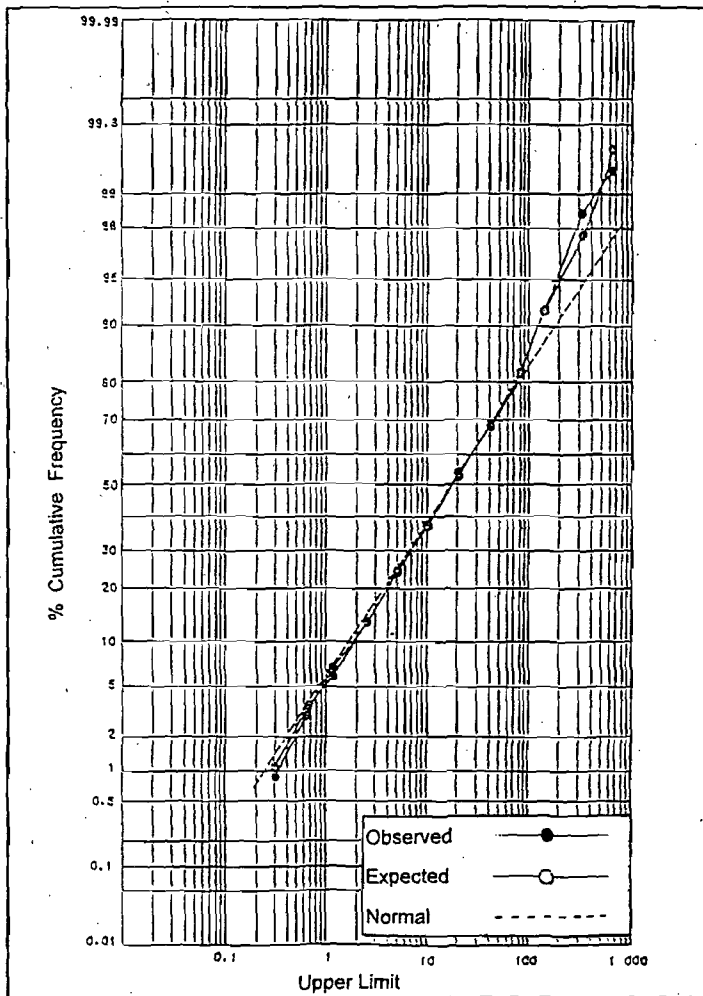


Figure 2. Cumulative log-probability plot showing sub-lognormality: Oil and Gas fields.

Plotting the observed $-\sqrt{\hat{\beta}_1}$ and $\hat{\beta}_2$ values on Figure 1, suggests initial estimates:

$$\gamma_0 \sim 0.46 \text{ and } b_0 \sim 0.70.$$

This yields solutions to (7) and (8) given by

$$\begin{aligned} \hat{\gamma} &= 0.46 & \text{and} & & \hat{b} &= 0.67 \\ \text{for which } \hat{\mu}'_1(u) &= 0.4487036, & & & \hat{\mu}_2(u) &= 0.9455061, \\ \hat{s} &= 1.6835482, & & & \hat{\xi} &= 2.038917, \\ x_M &= 3.1991278 & \text{and} & & n\hat{\varphi}(x_M) &= 156.05459. \end{aligned}$$

The expected frequency at x_M is the modal ordinate $n\hat{\varphi}(x_M)$ multiplied by the width of the class interval:

$$\ln(2) \cdot n\hat{\varphi}(x_M) = 108.16880$$

The expected value $z = e^x$

$$E(Z) = \int_0^{+\infty} z \Lambda(z) dz = e^{\xi} \frac{K_{\gamma+\xi}(b)}{K_{\gamma}(b)}, \quad (23)$$

where Λ is given in (19), and hence

$$\hat{E}(Z) = 50.896 \text{ MMBOE.}$$

The mode of the back-transformed distribution $\Lambda(z)$ is obtained from equation (21)

$$z_M = 0.7742 \text{ MMBOE.}$$

The modal ordinate for the $\Lambda(z)$ distribution is found from equation (22)

$$n\Lambda(z_m) = 44.77624.$$

The expected frequencies for the LNGIG model are based on equation (3) and the estimated $\hat{\gamma}$, $\hat{b}$, $\hat{s}$ and $\hat{\xi}$ values. The observed and expected frequencies are shown in table 4 below.

Table 4. LNGIG distribution fit to oil and gas field sizes

$x = \ln z$	Observed frequency ≥ 5	Expected frequency ≥ 5
-3.24 - -1.16	7	5.5
-1.16 - -0.47	15	12.9
-0.47 - 0.22	18	27.5
0.22 - 0.92	53	47.5
0.92 - 1.61	69	69.8
1.61 - 2.30	97	90.5
2.30 - 3.00	104	104.5
3.00 - 3.69	97	106.7
3.69 - 4.38	92	93.8
4.38 - 5.08	76	67.4
5.08 - 5.77	39	36.3
5.77 - 7.85	11	15.6
Total n	678	678.0

With a total χ^2 of 8.848 and $12 - 4 - 1 = 7$ degrees of freedom the associated probability is 0.264. Hence we cannot reject the hypothesis that the underlying distribution is LNGIG. The fit is illustrated in Figure 3.

7.2 Individual sample values from a hydrothermal gold ore deposit

The second example deals with individual gold ore values found in the Main Section of the Golden Valley Mine near Gatooma in Zimbabwe (Lightfoot, 1975). To quote from Lightfoot's thesis:

"The mine is a fairly typical hydrothermal gold-quartz vein occurring at the bottom contact of a laminated felsite in a greenstone schist belt on the edge of the 'Lion Hill' granite boss."

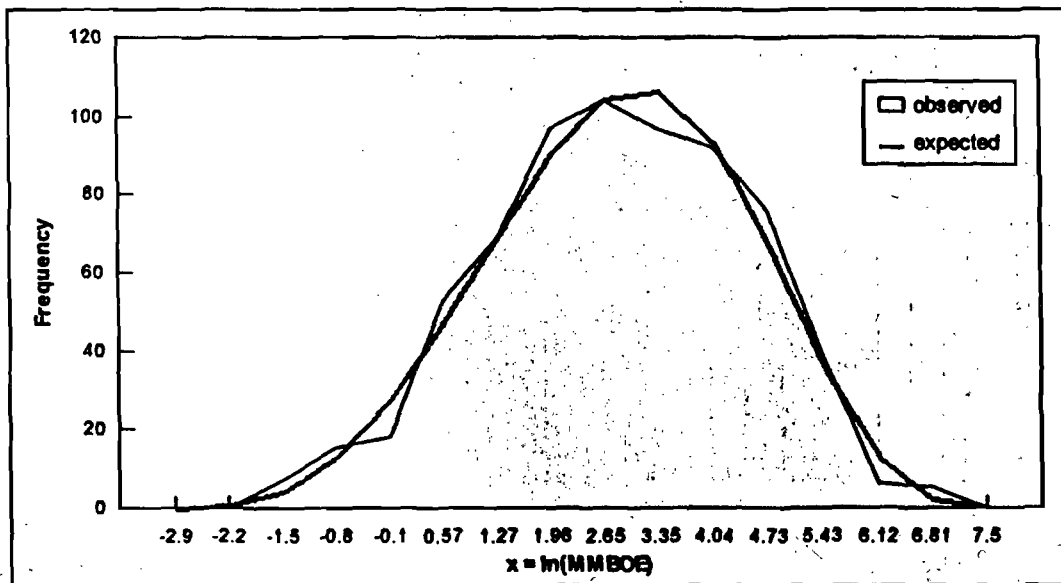


Figure 3. Frequency polygons for Oil and Gas Fields for the ordinates of $x = \ln(\text{MMBOE})$.

Table 5 gives the observed frequencies of 10 358 individual gold ore values in the Main Section of the Golden Valley Mine in terms of $x = \ln z$ class intervals.

Table 5. Observed frequency distribution of $x = \ln z$, for gold ore values in the Main Section of the Golden Valley Mine (after Lightfoot, 1975)

$x = \ln z$ class limits	Observed frequencies
0.446 - 1.139	4
1.139 - 1.833	30
1.833 - 2.526	90
2.526 - 3.219	259
3.219 - 3.912	602
3.912 - 4.605	1030
4.605 - 5.298	1576
5.298 - 5.991	1937
5.991 - 6.685	1870
6.685 - 7.313	1391
7.313 - 8.039	991
8.039 - 8.716	425
8.716 - 9.210	114
9.210 - 9.680	32
9.680 - 10.240	7
Total n	10358

The width of the classes are not all equidistant and steps were taken to take this into account when computing the first four central moments. These are $\hat{\mu}'_1(x) = 5.8261896$, $\hat{\mu}_2(x) = 2.0360137$, $\sqrt{\hat{\beta}_1(x)} = -0.1601726$ and $\hat{\beta}_2(x) = 2.817120$.

From the asymptotic test for normality of $\beta_2 = 3$ against $H_1: \beta_2 < 3$, we obtain:

$$\begin{aligned} \text{S.E. } [\hat{\beta}_2(x)] &= \sqrt{\frac{24}{n}} = 0.048136 \text{ and } t = \frac{2.817120 - 3}{0.048136} \\ &= -3.80 < -1.645 \end{aligned}$$

Thus we have shown that we have here a truly platykurtic distribution.

The logarithms of the observed frequencies of Table 5 are plotted against the midpoints of the logarithmic class intervals in Figure 4. The maximum is flat and the 'legs' curve inside towards the centre, thus corroborating visually that the underlying distribution is platykurtic.

Initial estimates for γ and b are obtained by plotting the observed skewness and kurtosis values on Figure 1, suggesting $\hat{\gamma}_0 \sim 1.0$ and $\hat{b}_0 \sim 2.6$.

This gives solutions to (7) and (8) given by

$$\hat{b} = 2.076 \quad \text{and} \quad \hat{\gamma} = 1.0,$$

$$\text{for which} \quad \hat{\mu}'_1(u) = 0.3947472 \quad \text{and} \quad \hat{\mu}_2(u) = 0.381531,$$

$$\hat{s} = 2.3100707 \quad \text{and} \quad \hat{\xi} = 4.9142956.$$

The observed and expected frequencies based on the estimates for γ , b , s and ξ given above are plotted in Figure 5, after a suitable adjustment for the unequal observed intervals. It can be seen that the agreement between the observed and expected frequencies for the LNGIG model is excellent.

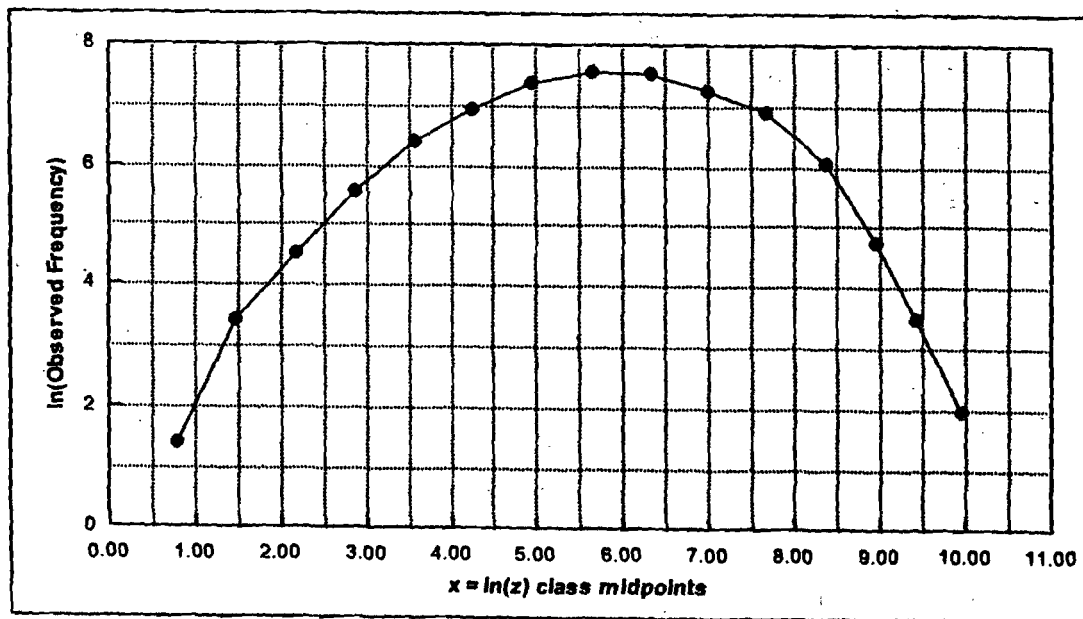


Figure 4. Golden Valley Gold Mine – log frequency plot.

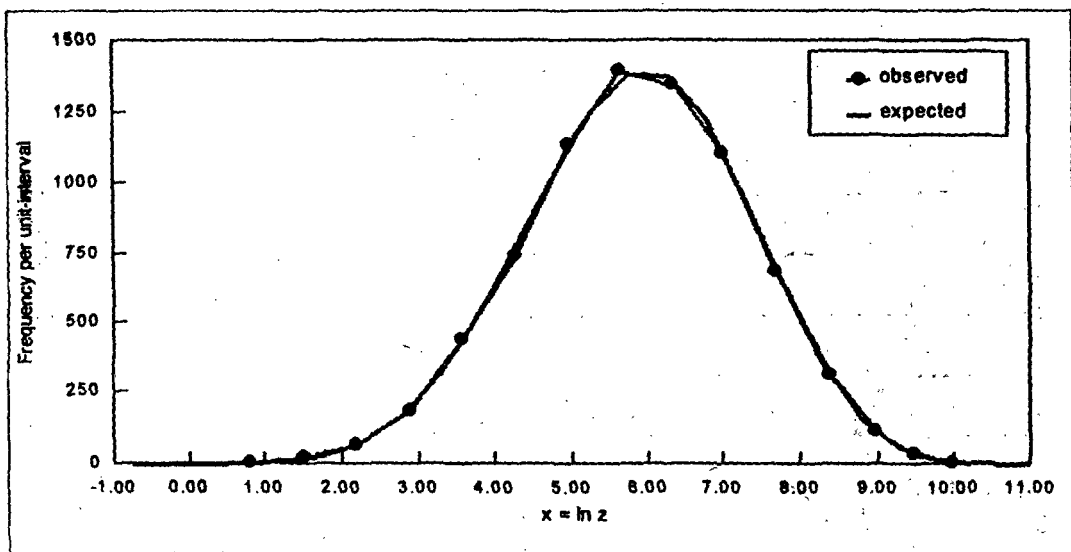


Figure 5. Golden Valley Mine – observed $x = \ln z$ distribution versus the fitted LNGIG model.

Finally, we back transform into the original values measured in inch-dwts. And obtain from

$$n\Lambda(z) = ne^{-x}\varphi(x) \\ 0 \leq z < \infty, \quad -\infty < x < \infty, \quad \text{and} \quad x = \ln z \quad (24)$$

that

1. the expected value of $\Lambda(z)$ based on $\hat{\gamma}, \hat{b}, \hat{s}$ and $\hat{\xi}$ is $E(\hat{Z}) = 850.4$ inch-dwts. (cf. the arithmetic mean of the data: $\bar{z} = 853.9$ inch-dwts.);
2. the mode for the back-transformed LNGIG, $n\hat{\Lambda}(z)$ is $z_M = 34.4385$ inch-dwts.
3. the modal ordinate for the $n\hat{\Lambda}(z)$ is given by: $n\hat{\Lambda}(z_M) = 24.3441$,
4. the expected frequency at this ordinate is $50 \cdot n\hat{\Lambda}(z_M) = 1217.207$. (50 inch-dwts. is the class interval width).

Expected frequencies are calculated for each z -class interval from the four-parameter GIG distribution given by equations (19). Table 6 shows the fit of the back-transformed LNGIG model to the observed frequencies as recorded by Lightfoot (1975).

Table 6. Observed frequency distribution of z (in inch-dwts.) fitted to the back-transformed LNGIG. Golden Valley Mine (Main Section).

z Class intervals (in inch dwts.)	Observed frequency	Expected frequency
0 - 50	985	1003.1
50 - 100	1030	1066.5
100 - 150	850	857.0
150 - 200	726	700.2
200 - 250	610	584.9
250 - 300	512	497.6
300 - 400	815	804.7
400 - 500	654	625.3
500 - 600	496	501.3
600 - 700	369	411.2
700 - 800	351	343.3
800 - 900	293	290.8
900 - 1100	444	464.9
1100 - 1300	392	353.7
1300 - 1500	262	276.7
1500 - 1700	212	221.1
1700 - 1900	170	179.7
1900 - 2100	146	148.2
2100 - 3100	463	458.0
3100 - 4100	227	222.6
4100 - 5100	123	122.2
5100 - 6100	75	72.6
6100 - 7100	54	45.7
7100 - 8100	29	30.0
8100 - 10000	31	33.4
10000 - 16000	32	33.6
>16000	7	9.7
Total n	10358	10358.0
χ^2		19.623
$p(\chi^2 df = 22)$		0.607

• Lightfoot's original histogram of the 10358 individual gold ore values and the back transformed LNGIG, are shown in Figure 6. Here again observations and theory agree.

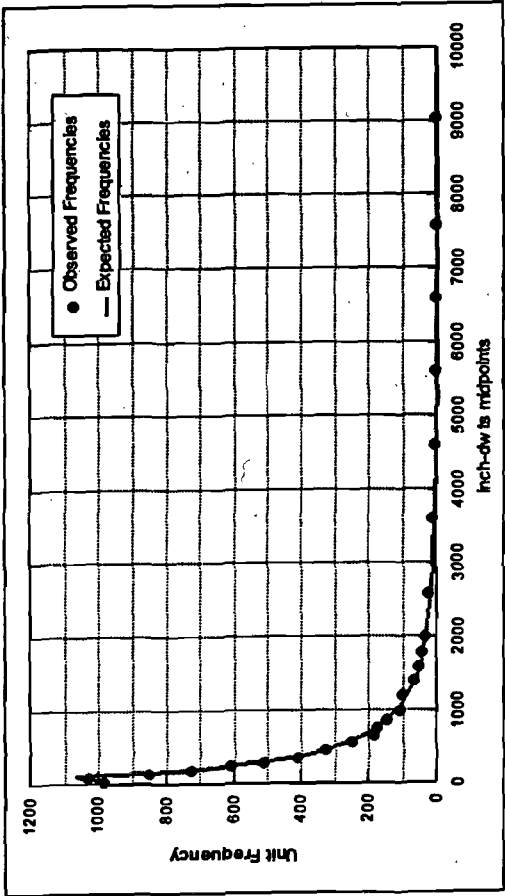


Figure 6. Golden Valley Mine (main section): 4-parameter GIG model fitted to observed individual ore values.

8. Conclusion

The LNGIG serves as a useful mathematical model for many kinds of observed data. This distribution has the remarkable property of short thin tails (i.e. its kurtosis $\beta_2 \leq 3$) in spite of its unlimited range at both ends of the real axis. Access to mathematical programming libraries which evaluate the K-function (e.g. Maple and IMSL) makes the application of the LNGIG model more attractive. In recent years this distribution has successfully been used to model the distribution of observed base metal and precious metal ore values, of which an example is provided.

9. Acknowledgement

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